THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — SCIENCE (086) · FULL SYLLABUS PRACTICE PAPER

Session 2026–27 | CBSE / RBSE Pattern | Section A Biology · Section B Chemistry · Section C Physics

Class: 10	Subject: Science	Set 05	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of these questions.
3. Draw neat, labelled diagrams wherever required. Use of calculator is not allowed.

SECTION A — BIOLOGY — 16 questions · 30 marks

Q1–Q7 MCQs. Q8–Q9: Assertion (A) / Reason (R) – choose A, B, C or D.

1. A student places a potted plant in a dark room for 48 hours and then tests a leaf for starch with iodine. The leaf turns: **1**

A. blue-black
B. colourless
C. brown/orange (no starch)
D. green

2. Which of the following is the correct sequence of the passage of urine? **1**

A. Kidney → ureter → urinary bladder → urethra
B. Kidney → urethra → bladder → ureter
C. Bladder → kidney → ureter → urethra
D. Ureter → kidney → urethra → bladder

3. A person touches a hot object and withdraws the hand without thinking. The nerve pathway of this action is: **1**

A. receptor → sensory neuron → spinal cord → motor neuron → effector
B. receptor → brain → motor neuron → effector
C. effector → motor neuron → spinal cord → receptor
D. receptor → motor neuron → brain → effector

4. Which hormone is released from the adrenal glands to prepare the body for ‘fight or flight’? **1**

A. Insulin B. Thyroxine C. Adrenaline D. Oestrogen

5. Spirogyra reproduces by breaking into smaller pieces, each of which grows into a new filament. This is: **1**

A. budding
B. fragmentation
C. spore formation
D. regeneration only

6. A plant with pure tall trait (TT) is crossed with a dwarf plant (tt). The F_2 generation shows tall and dwarf plants in the ratio: **1**

A. 1 : 1 B. 3 : 1 C. 2 : 1 D. 1 : 3

7. Non-biodegradable wastes like DDT enter the food chain and their concentration is highest in: **1**

A. producers
B. herbivores
C. primary carnivores
D. top carnivores

8. **Assertion (A):** The rate of breathing in aquatic organisms is much faster than in terrestrial organisms. **Reason (R):** The amount of dissolved oxygen in water is fairly low compared to the amount of oxygen in air. **1**

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

9. **Assertion (A):** Sexual reproduction leads to variation in a population. **1**
Reason (R): Variation in DNA copying during asexual reproduction produces no differences at all.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

10. List two functions of the human kidney other than the removal of nitrogenous wastes. Name the structural and functional unit of the kidney. **2**

11. Name the part of the brain that (i) controls balance of the body and (ii) controls the involuntary actions like blood pressure and salivation. **2**

OR

Why does the body need both nervous and hormonal control systems? Give one difference in their speed of action.

12. What is the role of the following in the human female reproductive system: (i) ovary, (ii) fallopian tube? **2**

13. Explain the process of double circulation in humans. Why is it necessary? **3**

14. Define biodegradable and non-biodegradable substances with two examples each. State one harmful effect of the accumulation of non-biodegradable waste. **3**

15. Priya's class performed an experiment: pollen grains were sprinkled on a sugar–boric acid solution on a glass slide and observed under a microscope after 30 minutes. They saw the pollen grains developing long tube-like structures.

(a) Name the structure formed by the pollen grain. 1

(b) Where does the pollen tube grow in a flower? 1

(c) Describe the process of fertilisation in flowering plants after the pollen tube reaches the ovule. 2

OR

(d) Distinguish between self-pollination and cross-pollination.

16. (i) Describe the structure of the human nephron with a labelled diagram. (ii) How is urine formed? (iii) What is the function of the artificial kidney? **5**

OR

(i) Describe the human male reproductive system with a labelled diagram. (ii) Write the functions of the testes. (iii) Name one sexually transmitted disease caused by a bacterium.

SECTION B — CHEMISTRY — 12 questions · 25 marks

Q17–Q21 MCQs. Q22: Assertion (A) / Reason (R) – choose A, B, C or D.

17. When carbon dioxide is passed through lime water, it turns milky due to the formation of: **1**

A. calcium oxide

B. calcium carbonate

C. calcium hydroxide

D. calcium bicarbonate

18. Which of the following salts does not contain water of crystallisation? **1**

A. Blue vitriol B. Washing soda

C. Common salt (NaCl) D. Gypsum

19. An element X forms an oxide that turns blue litmus red. The element X is most likely: **1**

A. a metal B. a non-metal

C. an alkali metal D. an alkaline-earth metal

20. Ionic compounds are generally soluble in water but not in kerosene because: **1**

A. water is polar B. kerosene is polar

C. both are non-polar D. ionic compounds are non-polar

21. Which of the following is a saturated hydrocarbon? **1**

A. C_2H_4 B. C_2H_2 C. C_3H_8 D. C_6H_6

22. **Assertion (A):** Hydrogen gas is not evolved when a metal reacts with nitric acid. **1**
Reason (R): HNO_3 is a strong oxidising agent and oxidises the H_2 produced to water.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

23. A metal ‘M’ on reacting with dilute sulphuric acid gives hydrogen gas. Write the balanced equation if M is zinc, and state whether M is above or below hydrogen in the reactivity series. **2**

24. Why are covalent compounds generally poor conductors of electricity? Give the reason for the low melting point of covalent compounds. **2**

25. Explain the term ‘pH’. Why is pH important in everyday life? Give two examples. **3**

OR

Write the chemical formula of bleaching powder. How is it prepared? State two uses.

26. Balance the following equations and identify the type of reaction: (a) $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$, (b) $H_2 + N_2 \rightarrow NH_3$, (c) $KClO_3 \rightarrow KCl + O_2$ **3**

27. A student takes three test tubes containing zinc granules. To the first she adds dilute HCl, to the second NaOH solution on heating and to the third distilled water. She observes bubbles of a colourless gas in the first two test tubes and none in the third. On bringing a burning splinter near the mouth of the first test tube the gas burns with a pop sound.

(a) Name the gas evolved in the first test tube. 1

(b) Write the balanced equation for the reaction in the first test tube. 1

(c) Write the name and formula of the salt formed in the second test tube. What does this reaction show about zinc? 2

OR

(d) Why is no gas evolved in the third test tube? Under what condition would zinc react with water?

28. (a) State the meaning of the term ‘homologous series’. Give two characteristics. (b) Name and write the formula of the first three members of the alkene series. (c) Write the electron-dot structure of ethene. **5**

OR

(a) What is the electrolytic process used for the manufacture of sodium hydroxide? Name the three products. (b) Write the balanced equation. (c) State one use of each product.

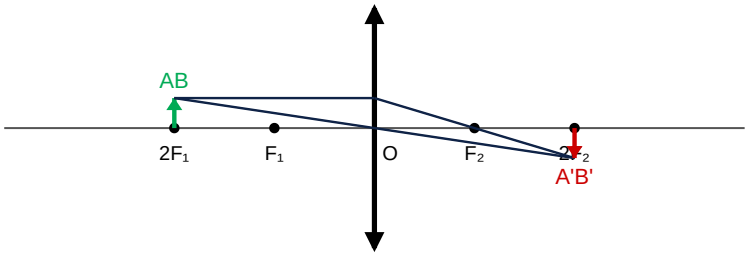
SECTION C — PHYSICS — 11 questions · 25 marks

Q29–Q32 MCQs. Q33: Assertion (A) / Reason (R) – choose A, B, C or D.

29. An object is placed at 40 cm in front of a convex mirror of focal length 20 cm. The image formed is at: **1**

A. 13.3 cm behind the mirror
B. 40 cm in front of the mirror
C. 13.3 cm in front of the mirror
D. 20 cm behind the mirror

30. The ray diagram shows an object placed in front of a convex lens of focal length 10 cm. The position of the image is: **1**



A. at 20 cm, real and inverted
B. at 15 cm, real and inverted
C. at 30 cm, virtual and erect
D. at 10 cm, real and inverted

31. A person with myopia cannot see distant objects clearly. The defect is corrected using a: **1**

A. concave lens
B. convex lens
C. cylindrical lens
D. bifocal lens only

32. The commercial unit of electrical energy is the kilowatt-hour. 1 kWh is equal to: **1**

A. 3.6×10^3 J B. 3.6×10^6 J C. 36×10^3 J D. 3.6×10^9 J

33. **Assertion (A):** A current-carrying straight conductor experiences no force when placed parallel to the magnetic field. **1**
Reason (R): The force on a current-carrying conductor is maximum when it is perpendicular to the magnetic field.

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

34. An object 5 cm tall is placed 25 cm from a concave mirror of focal length 10 cm. Find the position and nature of the image. **2**

OR

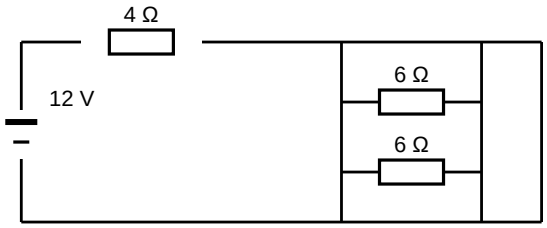
What is the cause of atmospheric refraction? Why do stars twinkle but planets do not?

35. Draw the magnetic field lines around a bar magnet. Mark the direction. What does the crowding of lines show? **3**

36. A wire of resistance 20 Ω is bent into a circular loop and then connected across a 10 V battery. What current flows? If the wire is stretched to double its length (volume constant), what is its new resistance? **3**

37. Explain why a concave mirror is used as a shaving mirror and a convex mirror as a rear-view mirror. **3**

38. In the figure, an electric circuit consists of a 12 V battery, an ammeter and a 4 Ω resistor in series with two 6 Ω resistors in parallel with each other. A student wants to find the current drawn from the battery and the potential difference across the resistor of 4 Ω.



(a) Calculate the equivalent resistance of the parallel combination. 1

(b) What is the total current drawn from the battery? 1

(c) Find the potential difference across the 4 Ω resistor. 2

OR

(d) Find the current through each 6 Ω resistor.

39. (a) State the laws of refraction. (b) A ray of light enters from air into glass of refractive index 1.5; find the speed of light in glass ($c = 3 \times 10^8$ m/s). (c) Draw a ray diagram of refraction through a rectangular glass slab. **5**

OR

(a) Describe the activity that shows a current-carrying conductor produces a magnetic field. (b) State the right-hand thumb rule. (c) How does the field change when the current is doubled?